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RESEARCH NOTE

Permo-Carboniferous (Kiaman) palaeointensity results from the western Bohemian Massif, Germany

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SUMMARY

Thellier palaeointensity analyses were carried out on samples from three quartz porphyry intrusions located within the western part of the Bohemian Massif close to the German–Czech Republic border. These intrusives have been dated at 280–275 Ma, which places them in the main part of the Permo-Carboniferous (Kiaman) Reversed Superchron (P-CRS). This age is consistent with the mean characteristic remanence direction calculated at $D/I = 204°/-1°$, giving a VGP at 36°N, 341°E. Palaeointensity values range from 6.6 μT to 22.1 μT, with a mean value of 12.0 ± 1.3 μT, which has a corresponding VDM value of $(3.0 \pm 1.2) \times 10^{22}$ A m^2. This suggests that the geomagnetic field strength recorded by the quartz porphyries is only 37 per cent of the current value, supporting recent studies that report a field strength of between 25 and 40 per cent of the present-day value during the P-CRS.

Key words: Europe, Kiaman superchron, palaeointensity, quartz porphyry.

1 INTRODUCTION

The strength of the Earth's magnetic field during periods of constant polarity (superchrons) is a topic of great current interest in palaeomagnetism. Recent studies (Pick & Tauxe 1994; Thomas, Rolph & Shaw 1995) have identified that the field intensity was biased towards values which range from 25 to 45 per cent of the present-day value during the Cretaceous Normal Superchron (CNS) and the Permo-Carboniferous (Kiaman) Reversed Superchron (P-CRS).

This paper documents a palaeointensity study of intrusive rocks from the western part of the Bohemian Massif, from near the German–Czech Republic border, which were emplaced during the P-CRS. Rock magnetic techniques are used to identify the remanence carriers and to assess the reliability of the palaeomagnetic record preserved by these rocks. The results are analysed in the context of a previous palaeomagnetic and rock magnetic study by Soffel & Harzer (1991) and compared to existing P-CRS palaeointensity data.

2 GEOLOGICAL SETTING AND SAMPLING

The metamorphic rocks of the Bohemian Massif straddle the border between the Czech Republic and western Germany and are separated from unmetamorphosed Palaeozoic–Mesozoic sediments by the Franconian line (Fig. 1). The rocks targeted for this study are small intrusions, which are surface expressions of large quartz porphyry bodies located in the western part of the Massif. These intrusions were emplaced into sandstones and siltstones between Upper Carboniferous and Lower Permian times and are regarded as being syngenetic or only slightly younger than the youngest granites of the Fichtelgebirge, which have been dated at 280 Ma by K/Ar and Rb/Sr methods (Besang *et al.* 1976). Soffel & Harzer (1991) quoted an age of between 275 and 280 Ma for the quartz porphyry bodies.

Sampling localities for this study were chosen in consultation with Prof. H. Soffel of the University of Munich, Germany, on the basis of a previous pilot palaeomagnetic study (Soffel & Harzer 1991). Of the 15 sites sampled in the pilot study, three were regarded as being suitable targets for a palaeointensity study, based on the rock type and stability of the NRM. Fig. 1 illustrates the locations of these sites. Between five and eight oriented (using a sun and magnetic compass) drill cores and several unoriented block samples were taken at each site.

3 PALAEODIRECTIONAL STUDY

The component structure of remanence for the quartz porphyries is well defined in the study of Soffel & Harzer (1991). They recognised that, after the removal of a component of

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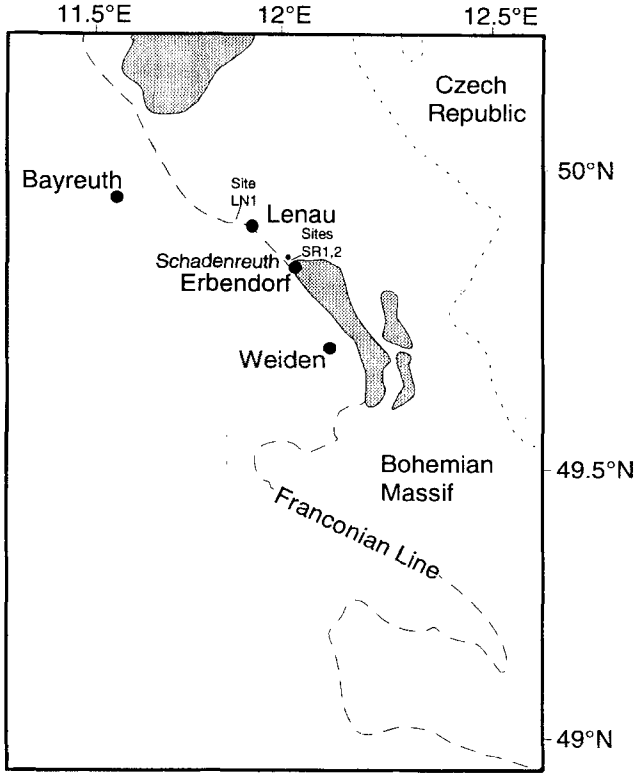


Figure 1. Map illustrating the sampling localities on the western margin of the Bohemian massif. Schadenreuth sites: SR1 = 49.8°N, 12.1°E; SR2 = 49.9°N, 12°E. Lenau site: LN1 = 49.9°N, 12°E. Shaded areas show the nappes of the Münchenberger Gneissmasse (northernmost shaded area) and the Zone-Erbendorf-Vohenstrauss (adapted from Soffel & Harzer 1991)

viscous origin below 200 °C, single straight-line segments converging on the origin were observed in orthogonal vector plots (OVPs). Maximum unblocking temperatures correspond to the Curie temperature of haematite, and no change in direction was observed after the Curie temperature of magnetite (580 °C), indicating that magnetite and haematite record the same remanence direction. The mean characteristic remanence direction for the quartz porphyries was calculated to be $D/I = 202.1/0.0$, $\alpha_{95} = 13.9$, $k = 80.1$.

Samples from each of the three sites targeted for the present study were stepwise thermally demagnetized and their remanence was measured using a Molspin fluxgate spinner magnetometer. Demagnetization and measurements were conducted inside a 2 m^3 Reubens' coil arrangement in order to minimize the effect of the ambient laboratory field. Samples were demagnetized in steps of 100 °C up to 300 °C and in 50 and 30 °C steps thereafter until the maximum unblocking temperature was reached.

Orthogonal vector plots of samples from each site are shown in Fig. 2 and site mean directions are listed in Table 1. Site mean directions obtained here are similar to those from equivalent sites in the previous study, and the OVPs confirm that magnetite and haematite generally record the same remanence directions. This suggests that these samples should be ideally suited for palaeointensity experiments since they are deuterically oxidized and have wide blocking-temperature spectra. The mean characteristic remanence direction for the intrusives ($D/I = 204.2/-1.0$, $\alpha_{95} = 6.6$, $k = 54.2$) is similar to that calculated by Soffel & Harzer (1991) and similar to the expected stable-field Permo-Carboniferous direction for central Europe, confirming the primary nature of the remanence.

Table 1. Palaeodirectional results for the quartz porphyry intrusives of the western Bohemian Massif. *D* and *I* are the mean declination and inclination of the characteristic remanent magnetization derived from the principal component analysis of *N* samples; $\alpha 95$ is the semi-angle of the 95 per cent cone of confidence; and *k* is the precision parameter. (a) Results from this study; (b) results from Soffel & Harzer (1991). The mean result for this study excludes the anomalous LN01 site, which has undergone post-cooling rotation.

Site	Mean direction					
		N	*D*	*I*	α_{95}	*k*
SR01	(a)	5	196.1	4.2	2.5	903.8
	(b)	23	197.1	8.5	2.9	107.0
SR02	(a)	5	213.8	−4.4	3.6	450.1
	(b)	16	206.6	−2.1	2.3	250.0
LN01	(a)	4	159.9	34.6	3.7	621.0
	(b)	26	162.1	34.9	3.1	82.1
Intrusive means	(a)	10	204.2	−1.0	6.6	54.2
	(b)	4	198.0	−2.5	13.8	45.1

4 MAGNETIC MINERALOGY

Soffel & Harzer (1991) performed IRM acquisition experiments, which indicated the presence of low and high blocking fields that they interpreted as being due to magnetite and haematite, respectively. Their reflected-light microscopy results confirmed that the magnetite and haematite were syngenetic, having been formed during a high-temperature oxidation and exsolution process. Therefore, they concluded that the remanence carried by both minerals is primary.

In the present study thermomagnetic and temperature-dependent susceptibility curves were employed, together with hysteresis loops and detailed IRM acquisition. The latter were used to model the sample blocking fields using an adaptation (Eyre 1994) of the methodology originally described by Robertson & France (1994), in which a log Gaussian distribution (or combination thereof for multiple magnetic phases) is fitted to the IRM acquisition data by an iterative process in which the variables that define each distribution are adjusted to minimize the discrepancy between the sample data and the curve produced by the model Gaussian distribution(s). The variables that define each distribution are the saturation (S) IRM of the phase, *B*1/2, the field at which half the SIRM (of that phase) is reached, and DP, the dispersion factor, which reflects the breadth of the blocking field of the phase (equivalent to one standard deviation, expressed as $\log_{10}$).

Thermomagnetic curves are shown in Fig. 3. Sample LN1-8 is dominated by a phase with a clearly defined Curie temperature close to 680 °C, consistent with haematite. There is also evidence of an inflection in the heating curve close to 580 °C, indicative of the presence of magnetite. At 100 °C the cooling curve shows a small reduction in magnetization by comparison with the heating curve, and the inflection at 580 °C is no longer evident. Sample SR2-7 also shows both magnetite and haematite Curie temperatures but the size of the haematite component is relatively much smaller. On cooling, a distinct magnetite Curie temperature is no longer visible and there is a reduction in the magnetization at 100 °C of approximately 15 per cent.

Sample SR1-HS shows similar behaviour, although the Curie temperature is closer to 550 °C, which indicates that Ti-poor titanomagnetite, rather than magnetite, provides the ferrimagnetic component.

Low-temperature susceptibility data are shown in Fig. 4. The broken lines show the data after correction for paramagnetic susceptibility. The paramagnetic susceptibility is determined from the high-field region of the hysteresis loops and will be an overestimate because of the presence of haematite. However, as is shown later, any such overestimate will be small for sample SR2-7. The low-temperature susceptibility curve for sample LN1-8 is essentially featureless, with no evidence of the Verwey transition (−150 °C) typical of multidomain magnetite or the Morin transition (−20 °C) seen in haematite. This behaviour is similar to that displayed by fine-grained synthetic magnetites (Senanayake & McElhinny 1981). Conversely, samples SR1-HS and SR2-7 show complex behaviour, both displaying two discrete local maxima in susceptibility. Sample SR1-HS has peaks centred at approximately −125 °C and −80 °C. The former is somewhat higher than the Verwey transition temperature, but the large sample size used could be responsible for a significant temperature lag between the interior of the sample and the thermocouple attached to the outside. The peak centred at −80 °C can be explained in terms of a Hopkinson peak for a low (−50 °C) Curie temperature phase. Candidates for such a phase are ilmenohaematites ($Fe_{2-X}Ti_XO_3$; −50 °C is equivalent to $X \approx 0.8$), which are common in igneous rocks. Slow cooling, such as will be experienced in large intrusives, leads to the exsolution of Ti-poor and Ti-rich phases from the original ilmenohaematite phase; the Curie curves have already identified the haematite component. Sample SR2-7 has peaks centred at approximately −150 and −60 °C, the former compatible with the presence of MD magnetite and the latter again consistent with a Hopkinson peak in an ilmenohaematite.

Hysteresis loops were determined using a Molspin vibrating sample magnetometer (VSM) with a peak field of ±1T (Fig. 5). The behaviour exhibited by samples LN1-8 and SR1-HS indicates the presence of high-coercivity (haematite) and low-coercivity (magnetite) phases. This is particularly clear for sample LN1-8, in which the two phases combine to produce a wasp-waisted hysteresis loop (Tauxe, Mullender & Pick 1996). The high-coercivity phase is not saturated by the available field, which is indicated by loops that are open at high fields. The characteristics of sample SR2-7 are less clear because of the dominance of an apparently paramagnetic component (the presence of a phase with coercivities that exceed 1 T will contribute to the high-field behaviour). Subtraction of this component (shown by the broken line) confirms that the remanence is dominated by a low-coercivity titanomagnetite phase.

The IRM acquisition curves and subsequent modelling of the data proved to be the most useful rock magnetic technique. In Fig. 6(a) the normalized IRM curves clearly indicate the presence of low and high blocking fields in all three samples. This is emphasized in Figs 6(b)–(d); here the data points indicate the 'gradient' (strictly the derivative of the IRM acquisition curve) between pairs of points shown in Fig. 6(a) (plotted against the mid-field value for each pair) and the bell-shaped curves represent the best-fit log-Gaussian distribution for the two identified phases. The height, width and centre field of each distribution have been chosen such that the cumulative log-Gaussian curve, indicated by the thicker line, provides an acceptable fit to the data points. The values assigned to each of these fitting parameters are indicated on the figures. All three samples show the presence of two discrete distributions, one low blocking field ($B1/2$ 33–56 mT) and one high blocking field ($B1/2$ 500–660 mT); the $B1/2$ values of the low blocking field distribution are compatible with magnetite in the small MD/PSD to SD grain-size range, while the high blocking field component has $B1/2$ values typical of haematite (Dunlop 1972). The IRM data therefore confirm the identification of (titano)magnetite and haematite determined from the Curie curves. Furthermore, they indicate a mean magnetite grain size in the PSD size range, which is encouraging for the palaeointensity study. In samples LN1 and SR1 the SIRM is split 40/60 between the two phases; in sample SR2 almost 90 per cent of the SIRM resides in the (Ti-poor titano)magnetite phase, confirming the hysteresis data.

5 PALAEOINTENSITY ANALYSES

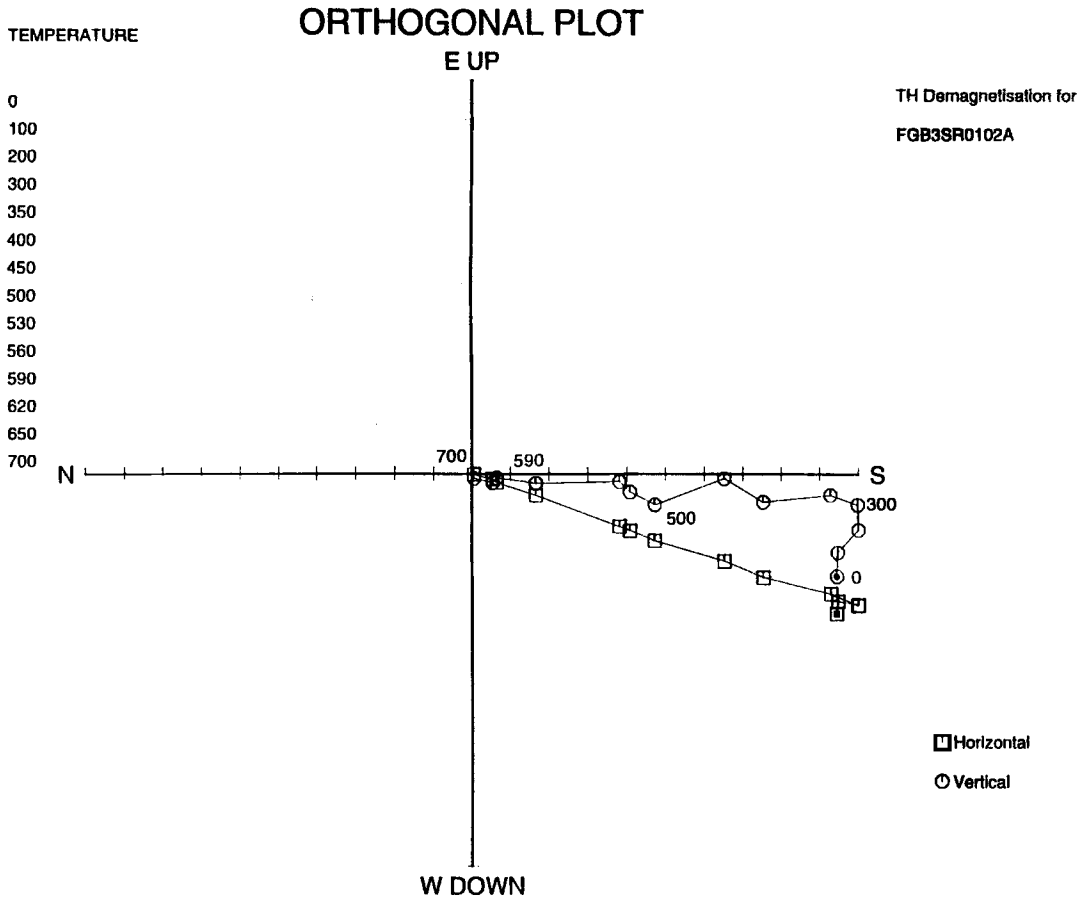
Palaeointensity analyses were performed using a modified version of the Thellier (Thellier & Thellier 1959) technique, similar to that used by Prévot *et al.* (1985), which incorporates PTRM checks at 100 °C intervals to check for non-linear alteration during laboratory heating. A laboratory field of 25 μT was used to impart TRMs and the temperature steps used in the experiments were chosen with reference to the results of thermal demagnetization (section 3) in order that the most appropriate part of the blocking-temperature spectrum could be used to define the palaeointensity result. Room-temperature susceptibility variations were monitored at each demagnetization step during Thellier experiments in order to provide a further check for alteration and creation/destruction of mineral phases.

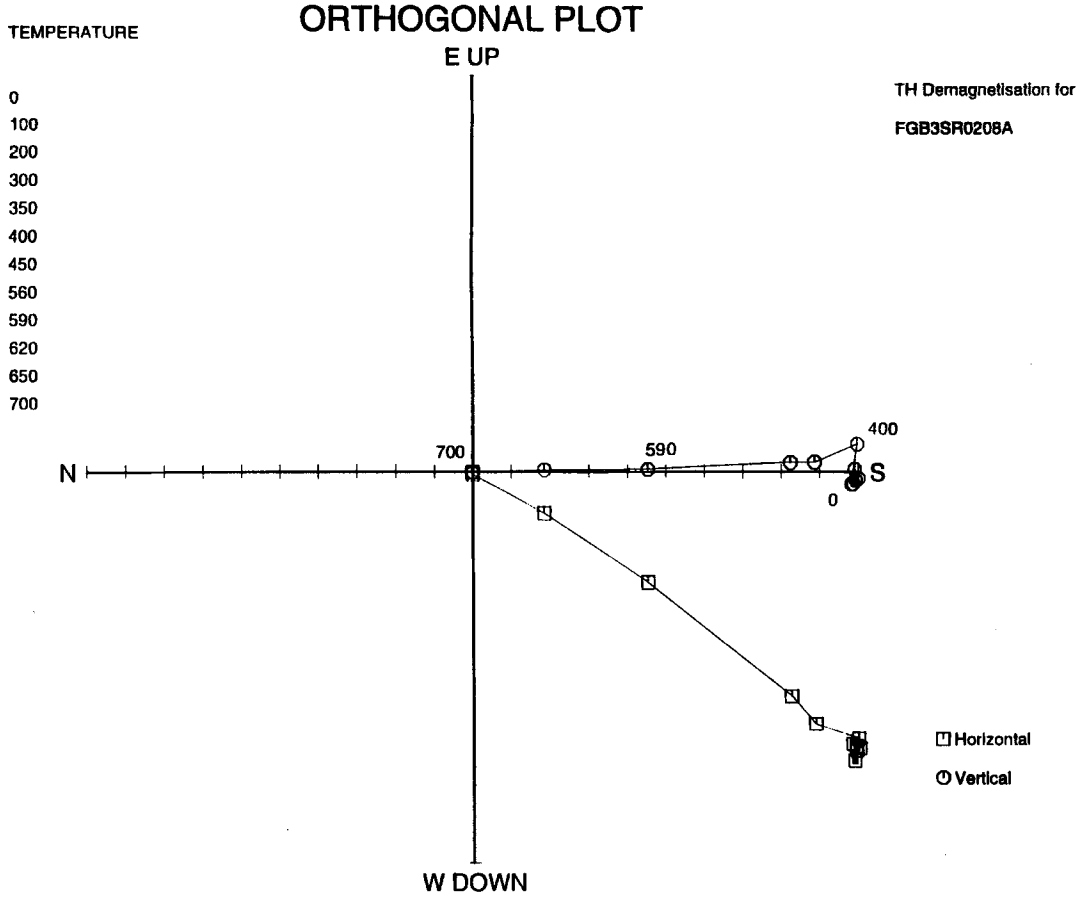
Results of palaeointensity experiments are listed in Table 2 and include the accepted standard statistical parameters used by Coe, Grommé & Mankinen (1978). Fig. 7 illustrates typical NRM–TRM (Arai) plots for samples from each site.

The high degree of oxidation to which the intrusives have been subjected allows the palaeointensity to be defined over a larger blocking temperature interval than is usually possible for ancient igneous rocks. Alteration tends to set in above 600 °C for most samples, whereas this commonly occurs at temperatures of 450–500 °C in most ancient igneous rocks. PTRM checks are deemed to be successful if the PTRM value recorded at a particular temperature and its check value are within 15 per cent of each other. PTRM checks show a reasonable success rate when compared to studies of other ancient igneous rocks (e.g. Thomas *et al.* 1995) and, as is common, highest success rates are observed for PTRM checks at 300 and 400 °C and, less commonly, at 500 °C (Fig. 8). The palaeointensity values recorded range from 6.6 μT to 22.1 μT, with an overall mean value of 12.0±1.3 μT, corresponding to a virtual dipole moment (VDM; Smith 1967) of $(3.0 \pm 1.2) \times 10^{22}$ A m^2.

6 DISCUSSION AND CONCLUSIONS

The rock-magnetic work of Soffel & Harzer (1991) and the present study have shown that the magnetic mineralogy of the quartz porphyry intrusions consists of magnetite and haematite in varying proportions. Grommé, Wright & Peck (1969)


ORTHOGONAL PLOT
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TH Demagnetisation for
FGB3SR0102A
Horizontal
Vertical
W DOWN


ORTHOGONAL PLOT
TEMPERATURE
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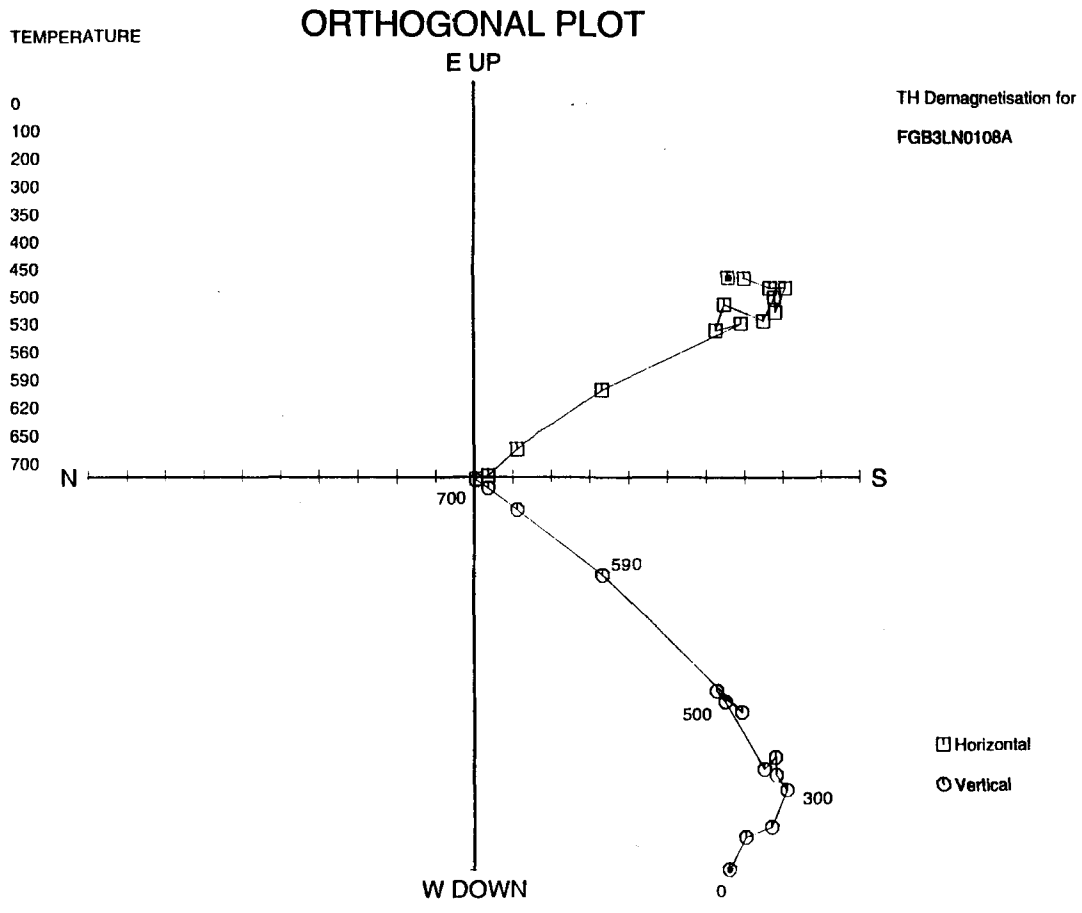


Figure 2. Representative results of palaeodirectional analysis for the quartz porphyries, plotted as orthogonal vector (Zijderveld) plots (OVPs). Open circles represent projections onto the vertical plane and open squares represent projections onto the horizontal plane. Temperature steps are shown (°C).

recognized that it is possible for deuteric oxidation to continue down to temperatures as low as 300 °C in rapidly cooled igneous rocks where the oxidation reactions lag behind the temperature as the igneous body cools. This is unlikely to occur, however, in the quartz porphyries, where cooling is very much slower, and therefore the rocks have sufficient time to oxidize at high temperatures. This suggests that oxidation of original titanomagnetites occurred at high temperatures during initial cooling. The samples are therefore ideally suited to Thellier palaeointensity studies. This allows the palaeointensity to be defined over a much wider blocking-temperature range than is normally possible in studies of ancient rocks. This definition over a wide range also makes it unlikely that the low palaeointensities witnessed are due to the spontaneous decay of NRM, which is sometimes a problem in ancient rocks. The characteristic increase in PTRM capacity commonly observed at around 500 °C (e.g. Thomas *et al.* 1995) is not witnessed until after 600 °C here.

The differences in characteristic mean remanence direction, mean palaeointensity and magnetic mineralogy observed between sites SR1 and SR2 can be explained in a number of ways. These sites may be taken from different parts of the same intrusion, separated by some 2 km, and differences in the field may be due to palaeosecular variation (PSV). The coarse-grained nature of the intrusions, suggesting slow cooling, supports this explanation. Alternatively, the sites may have been taken from different, and hence time-distinct, intrusions, a view that receives some support from the fact that the magnetic mineralogy of the two sites is dominated by mineral phases with markedly different blocking fields. The lack of exposure of the intrusion between sites SR1 and SR2 precludes confirmation of this, although the raised topography between the two sites suggests that they may be part of a single body. If this is true, then SR1 and SR2 may well record different pulses of igneous activity, the expression of which results in a single igneous intrusion.

The possibility of a later hydrothermal event causing overprinting of one site in preference to another is unlikely. Thermal demagnetization results confirm only a single component of remanence through the blocking-temperature spectra of magnetite and haematite for both SR1 and SR2. There is no geological support for remagnetization up to 700 °C, and rock magnetic studies show no evidence of the existence of maghemite, which would result from low-temperature oxidation. A more plausible explanation for the difference observed is that one of the sites has been rotated slightly with respect to the other. It is not difficult to see that this could have occurred, because the site at Lenau (LN1) is known to have undergone some rotation (Soffel & Harzer 1991), which is confirmed by its anomalous site mean characteristic remanence direction. This explanation would not, however, account for the differences in palaeointensity values or magnetic mineralogy.

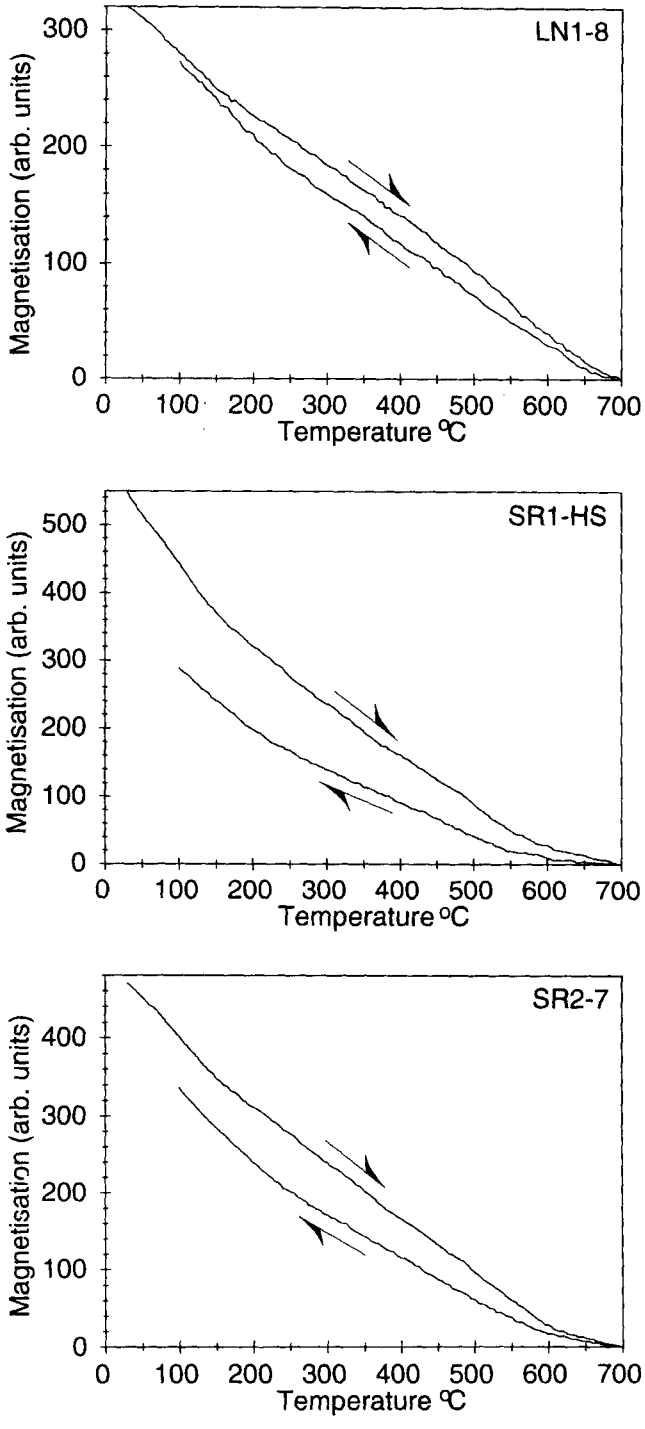


Figure 3. Representative thermomagnetic (*Js* versus *T*) curves for the quartz porphyries.

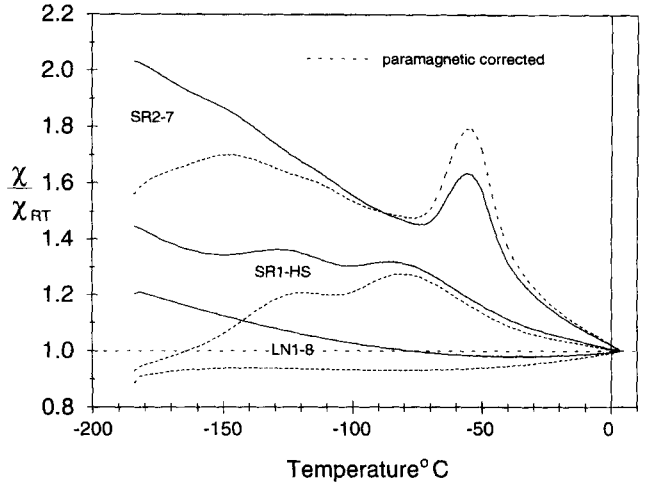


Figure 4. Representative low-temperature susceptibility curves (χ/χ_{RT}) for the quartz porphyries. The horizontal dashed line is the $\chi/\chi_{RT}=1$ line.

Furthermore, we do not have enough data to produce a confident distinction between PSV and rotation as possible explanations.

The slow-cooled nature of the intrusions will affect the palaeointensity results, but a detailed analysis of this effect is precluded by a number of factors, including our uncertainty of the size of the intrusions and the inability to demonstrate the exact position of our samples within the intrusive body. The important factor for cooling-rate correction is the order of magnitude difference between the laboratory and natural rates. For the slow-cooled intrusives we can estimate that the time taken for our samples to cool through their blocking temperatures would be somewhere between 50 and 50 000 yr. This equates to a decrease in the cooling rate of between 6 and 9 orders of magnitude when compared to a laboratory cooling field of around an hour. According to Fox & Aitken (1980) and Dodson & McClelland-Brown (1980) this would lead to a 40–60 per cent increase in TRM and an equivalent overestimate of palaeointensity. Such approximations illustrate the difficulty in attempting to quantify a cooling-rate-corrected palaeointensity value. However, the important point is that any such effect would result in an overestimate of the palaeointensity, which merely reinforces our main argument, namely that the dipole field was much weaker during this part of the Kiaman than the present-day value.

Whereas the palaeodirectional results presented here do not add significantly to the analysis of Soffel & Harzer (1991), the importance of the palaeointensity results should not be underestimated. The quartz porphyries are highly suitable for palaeointensity work, and their behaviour during experiments confirms this suitability. Furthermore, the palaeodirections correspond to the expected Permo-Carboniferous direction for central Europe. We are therefore confident that the low palaeointensity recorded is a reliable, stable-polarity estimate and provides further evidence to suggest that the strength of the geomagnetic field was low during the Kiaman reversed superchron. The mean VDM value for the quartz porphyries, plotted relative to the present-day mean VDM of 8.02×10^{22} A m^2, is close to the values for the 300 Ma Midland Valley Sill of Scotland (Thomas *et al.* 1995) and the 280 Ma Mauchline lavas of Scotland (Harcombe-Smee *et al.* 1996). The results of the present study indicate two (possibly three) small, slow-cooled igneous intrusions that are highly suitable for palaeointensity work and thus document an additional data point on the P-CRS palaeointensity curve. No elaborate discussion of the causes of the low field intensity can be made from the results of the present study in isolation.

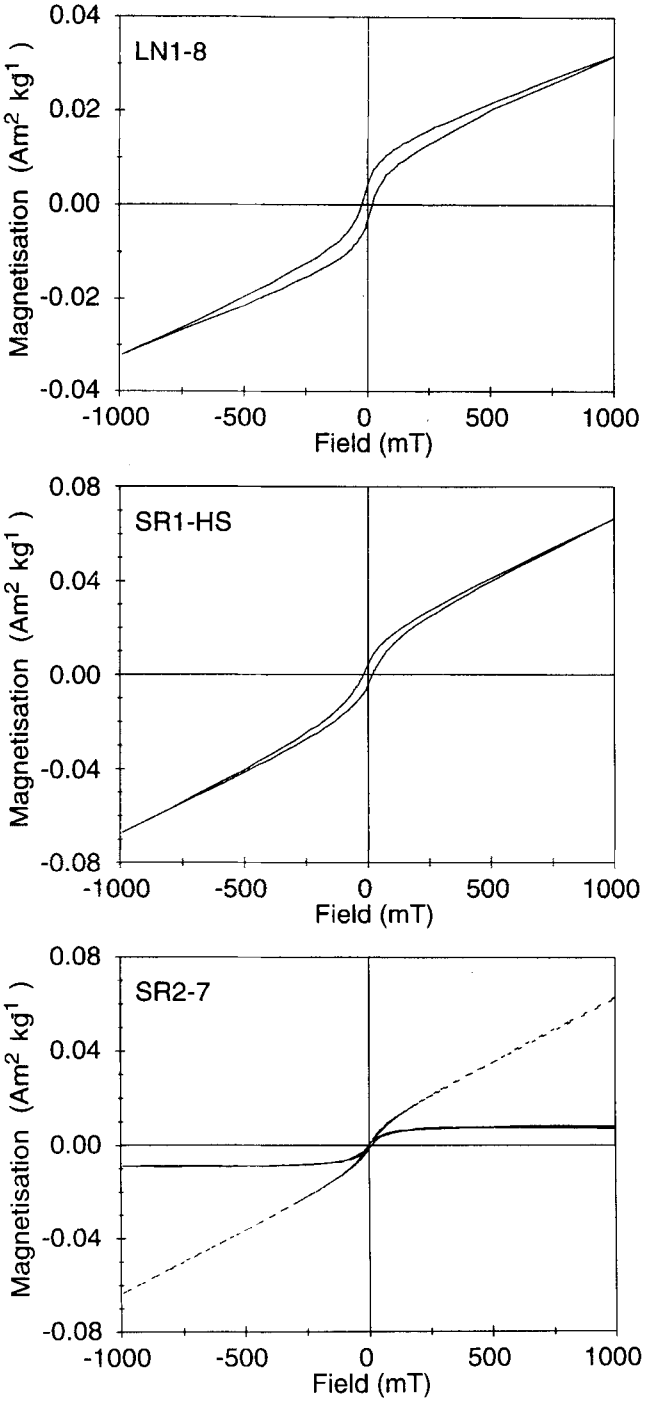


Figure 5. Representative hysteresis curves for the quartz porphyries. The broken curve in sample SR2-7 shows the total (paramagnetic + ferrimagnetic) components.

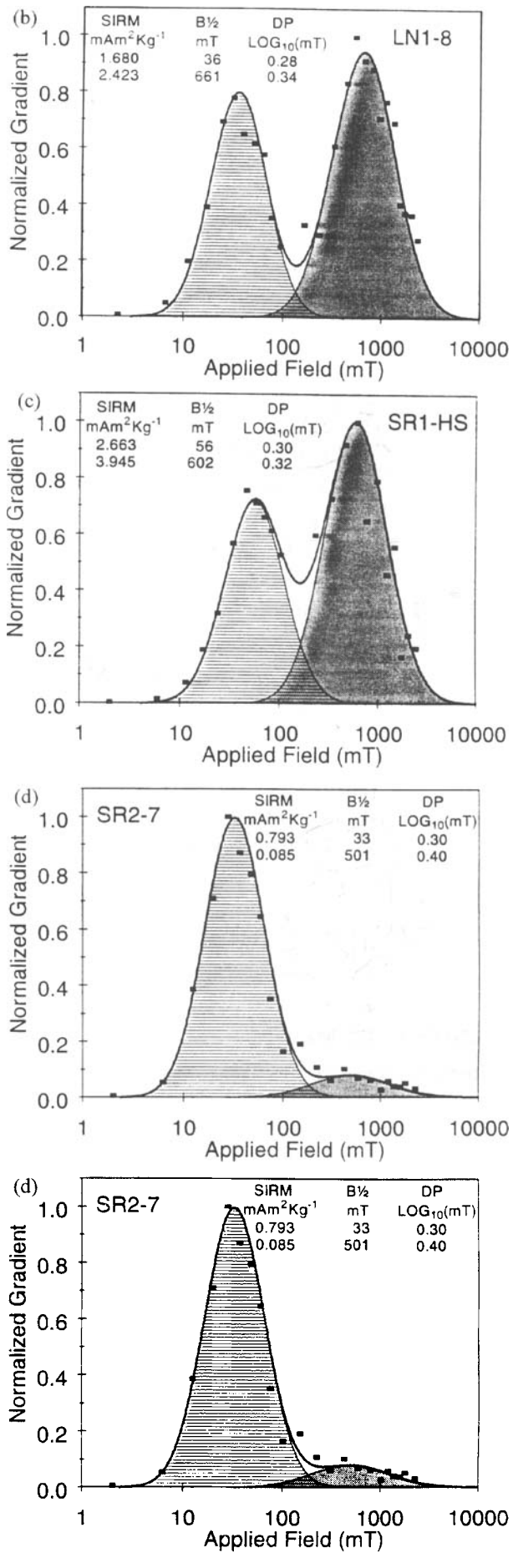


Figure 6. (a) Normalized IRM acquisition curves for the quartz porphyries. (b)–(d) IRM acquisition curves expressed as a gradient with best-fitting coercivity distributions shown as line fits to the data. Solid squares are the data points, thin lines show the fit of individual phases, thick lines show the combined fit of the phases. SIRM = saturation IRM of the phase, $B1/2$ = field at which half the SIRM (of that phase) is reached and DP = the dispersion factor reflecting the breadth of the coercivity distribution of that phase (equivalent to 1 s.d., expressed as $\log_{10}$).

Table 2. Thellier palaeointensity results for the quartz porphyry intrusions of the western Bohemian massif. *T* range: the temperature range over which *Ba* was calculated; *N*: number of data points used to calculate *Ba*; *f*, *g* and *q* are the NRM fraction, gap factor and quality factor, respectively, of Coe *et al.* (1978); *Ba*: the Thellier palaeointensity result in μT; VDM: the mean virtual dipole moment for each site ($\times 10^{22}$ A m^2).

Sample	*N*	*T* range	*f*	*g*	*q*	*Ba*	VDM
SR01-02B	8	450–660	0.949	0.768	10.10	8.1±0.6	
SR01-03B	9	400–660	0.907	0.767	12.19	6.6±0.4	
SR01-04B	9	400–660	0.893	0.770	16.14	7.0±0.3	
SR01-05B	8	400–630	0.837	0.740	16.57	7.5±0.3	
Mean						7.3±0.6	1.9±0.2
SR02-01B	9	300–630	0.934	0.754	6.86	14.0±1.4	
SR02-03B	8	400–630	0.840	0.767	4.90	18.9±2.5	
SR02-06B	10	250–630	0.917	0.709	5.60	17.0±2.0	
SR02-08B	10	250–630	0.931	0.778	17.33	22.1±0.9	
Mean						18.0±3.4	4.6±0.9
LN01-02B	9	200–660	0.972	0.333	6.39	7.6±0.4	
LN01-03B	11	250–660	0.904	0.790	14.64	9.3±0.5	
LN01-05B	6	200–450	0.264	0.300	0.27	11.5±3.4	
LN01-07B	5	250–450	0.191	0.655	4.33	14.4±0.4	
Mean						10.7±2.9	2.4±0.7
Overall mean		3 sites				12.0±1.3	3.0±1.2

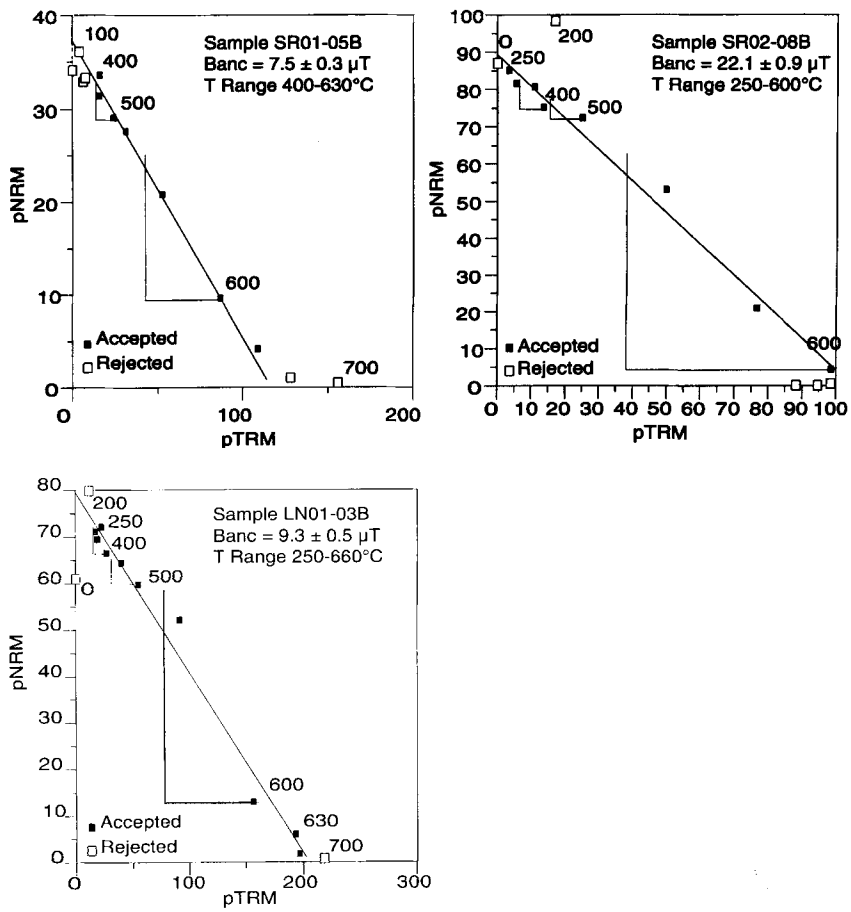


Figure 7. Representative Arai (pNRM–pTRM) plots from Thellier–Thellier palaeointensity experiments for the quartz porphyries. Solid (open) squares denote accepted (rejected) points from the best-fit line through the palaeointensity data. Banc is the calculated palaeointensity value; T Range is the temperature range over which Ba is calculated; PRTM checks are taken at elevated temperatures.

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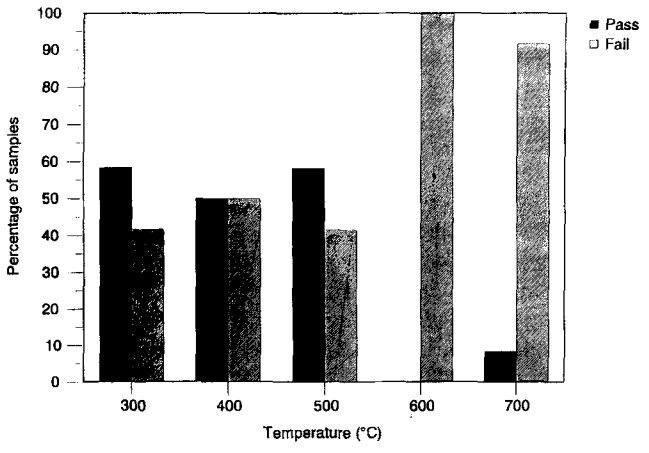


Figure 8. Histograms showing pass/fail percentages for samples during PTRM checks at 300, 400, 500, 600 and 700 °C.

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